SL

CAPSTONE PROJECT

StockLink

Hyper-local B2B wholesale marketplace & real-time mandi price discovery

TEAM

Praween Singh Rawat · Anvitha S · Ashraj Singh Rana · Tanishq Mittal

Front End Frameworks and Technologies · MCA (General), Semester I

Alliance University, Bengaluru · Faculty: Veera Raghav K

Agenda

1. What StockLink does, with a live demo
2. Technologies and team responsibilities
3. Advantages, disadvantages and future scope
4. Hosting link and our learning

What StockLink does

THE PROBLEM

Traditional mandi wholesale procurement relies on fragmented phone calls, physical visits, and opaque broker pricing.

Kirana retailers overpay 12–18% in commission markups and face frequent stockouts due to phantom/ghost inventory.

HOW IT WORKS

1 Wholesalers list verified stock with live unit counts and editable base prices.

2 Retailers discover local mills by distance and lock rates below APMC mandi benchmarks.

3 Direct procurement orders are placed with verified GSTIN credentials and tracked in real time.

12.8% Below APMC Mandi Rate — Basmati Rice ₹85/kg

Official Mandi Benchmark: ₹95/kg · Direct Mill Rate: ₹85/kg (Save ₹10/kg)

LIVE MARKETPLACE DEMONSTRATION

✓ Retailer Mandi Discovery: Active suppliers count (12 verified mills) within 5km radius with real-time stock units.

✓ Direct Procurement Cart: Source Cart locks quantities with bulk discount thresholds and automated GST tax calculations.

✓ Wholesaler Inline Pricing: Live SKU base price edits update instantly across connected buyer screens without reload.

✓ Fulfilment Queue: Active tracking: Order #SL-7401 dispatched with verified GSTIN trader credentials.

A live session on the hosted app · Live demo · fft-capstone-project.vercel.app

Technologies used & team responsibilities

TECHNOLOGIES

Front end

React 18 · Vite 6 · React Router · Tailwind CSS · Lucide Icons

Real-time backend

Node.js · Express.js · REST APIs · JWT Auth Middleware

Database & Storage

MongoDB Atlas · Mongoose ORM · Resilient In-Memory Dual Store

Testing and hosting

Vercel (Edge CDN) · Render (Cloud Web Service) · Automated Suite

TEAM RESPONSIBILITIES

[PR] Praween Singh Rawat

FRONTEND MAIN LEAD

UI/UX architecture & design tokens; interactive Retailer & Wholesaler Dashboards; comprehensive profile management engines; responsive layouts.

[AS] Anvitha S

FRONTEND DEVELOPER

Multi-role authentication workflows; 3-stage onboarding wizard (GSTIN, Address, Preferences); centralized API service (`api.js`); Source Cart state.

[AR] Ashraj Singh Rana

BACKEND MAIN LEAD & DEVOPS

Express REST API architecture; real-time dashboard aggregation engines (spend, revenue, APMC benchmark %); cloud deployment on Render & Vercel.

[TM] Tanishq Mittal

BACKEND & DATABASE DEVELOPER

Database connection & Mongoose schemas (User, Product, Order); role-based JWT security middleware; resilient in-memory dual fallback engine.

Advantages, disadvantages & future scope

Advantages

- **Direct disintermediation**
  Direct mill-to-retailer linkage eliminates 12–18% middleman commission markups.
- **Zero ghost inventory**
  Real-time stock reservation prevents cancelled orders and stockouts.
- **Live APMC price discovery**
  Empowers local shopkeepers with live mandi benchmark pricing.
- **Resilient dual engine**
  In-memory fallback guarantees 100% platform uptime during database outages.

Disadvantages

- **Every trader needs connectivity**
  Live inventory counts and rate updates require continuous internet access.
- **Local geofence boundaries**
  Logistics radius is currently tailored for localized mandi hubs (<15–20 km).
- **Self-managed logistics**
  Delivery relies on existing local transporter fleets without automated route tracking.
- **Single currency ecosystem**
  Transactions are optimized exclusively for Indian Rupees (INR) and GSTIN formats.

Future enhancements

- **AI predictive procurement**
  ML forecasting of mandi price trends and automated Kirana stock replenishment.
- **Automated e-Way bills**
  Instant GST e-Way bill and tax invoice generation upon order dispatch.
- **Integrated escrow gateway**
  Milestone-based digital payment holding until delivery confirmation by retailer.
- **Vernacular voice search**
  Multilingual voice search (Hindi, Kannada, Telugu, Marathi) for local mandi traders.

Hosting link & our learning experience

LIVE AT

fft-capstone-project.vercel.app



Scan to try it

Join as a retailer or sign in as a wholesaler with pre-configured demo credentials.

Runs on Vercel Edge Network (frontend) and Render Cloud Web Service (Node.js backend).

WHAT WE LEARNED

1 Thinking in components: A real React 18 application with hooks, client routing and reactive cart state.

2 Full-stack API orchestration: Live sync with Express REST APIs, centralized controllers and CORS rules.

3 Role-based security & RBAC: Stateless JWT authorization, password hashing with bcryptjs, and route protection.

4 Resilient data persistence: Designing multi-tier storage with MongoDB Mongoose schemas and an automatic in-memory fallback.

5 Shipping production software: Configuring edge CDN routing, SPA rewrites, automated test suites and zero-downtime cloud hosting.

Thank you · Questions?